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MULTI-OBJECTIVE SIZING FOR HYBRID SOLAR SYSTEMS USING SYNTHETIC RESIDENTIAL LOADS AND MACHINE-LEARNING-BASED LONG-TERM IRRADIANCE-TEMPERATURE FORECASTS

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Research proposal for a thesis to be submitted in partial fulfilment of the requirements for the award of the degree of Master of Science in Mechatronic Engineering of Jomo Kenyatta University of Agriculture and Technology

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DECLARATION

This research proposal is my original work and to the best of my knowledge it has not been presented for a degree award in this or any other university.

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NOMENCLATURE

Symbol	Description
P_{PV}	PV array capacity (kW)
E_{bat}	Battery storage capacity (kWh)
P_{inv}	Hybrid inverter rating (kW)
P_{bat}	Battery power rating (kW)
G_t	Solar irradiance (W/m^2)
T_t	Air temperature (°C)
L_t	Household load demand (kW)
P_t^{PV}	PV generation at time t (kW)
P_t^c	Battery charging power at time t (kW)
P_t^d	Battery discharging power at time t (kW)
P_t^{grid}	Grid import/export power at time t (kW)
E_t^{ns}	Energy not supplied at time t (kWh)
E_t^{served}	Energy served by PV+battery at time t (kWh)
SOC_t	Battery state of charge at time t
u_t^c	Battery charge switch (0/1)
u_t^d	Battery discharge switch (0/1)
w_t	Representative day weight
c_{PV}	PV capital cost (KES/kW)
c_{bat}	Battery capital cost (KES/kWh)
c_{inv}	Inverter capital cost (KES/kW)
c_{BOS}	Balance of system cost (KES)
G_{STC}	Irradiance at standard test conditions (1000 W/m^2)
T_{STC}	Temperature at standard test conditions (25°C)
γ	PV temperature coefficient (°C^{-1})
η_c	Battery charge efficiency
η_d	Battery discharge efficiency
DOD	Depth of discharge
D	Days of autonomy
$\bar{L}_{\text{daily}}$	Weighted average daily load (kWh/day)
n_t	Active residents at time t
$p_{i,t}$	Appliance i switch-on probability at time t
$P(n_t \mid n_{t-1})$	Markov occupancy transition probability
$\text{TOU}_{i,t}$	Time-of-use probability for appliance i at time t
$p_{\text{light},t}$	Lighting switch-on probability at time t
irradiance_t	Irradiance at time t for lighting model
bulbs	Number of lighting bulbs
CSI_t	Clear Sky Index at time t
$\text{GHI}_{\text{clear-sky},t}$	Clear sky GHI at time t
T_L	Linke turbidity coefficient

Symbol	Description
$\mathbf{h}_t$	Hidden state at time t (neural networks)
$\mathbf{X}_{t-11:t}$	Input window (12 months)
z_t	GRU update gate
$\tilde{\mathbf{h}}_t$	GRU candidate activation
$\hat{y}_m$	Forecasted value for month m
σ_m	Forecast uncertainty for month m
$f_1(x)$	Cost objective function
$f_2(x)$	Self-sufficiency objective function
ϵ_i	ε-constraint level i
$\mathcal{P}$	Pareto front
σ_{model}	Model standard deviation
σ_{meas}	Measured standard deviation
$L_{\text{model},t}$	Model load at time t
$L_{\text{meas},t}$	Measured load at time t
Δt	Time step

ABBREVIATIONS

Abbreviation	Full Form
BESS	Battery Energy Storage System
CNN	Convolutional Neural Network
CREST	Centre for Renewable Energy and Sustainable Technologies
CSI	Clear Sky Index
DOD	Depth of Discharge
GHI	Global Horizontal Irradiance
GRU	Gated Recurrent Unit
HRES	Hybrid Renewable Energy System
IRR	Internal Rate of Return
KES	Kenyan Shilling
KPLC	Kenya Power and Lighting Company
LCOE	Levelized Cost of Energy
LPSP	Loss of Power Supply Probability
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MECS	Modern Energy Cooking Services
MECS-CREST	MECS Modified CREST Model
MILP	Mixed Integer Linear Programming
MC	Monte Carlo
NASA POWER	NASA Prediction of Worldwide Energy Resources
NPC	Net Present Cost
NSGA-II	Non-dominated Sorting Genetic Algorithm II
O&M	Operation and Maintenance
PV	Photovoltaic
PVGIS	Photovoltaic Geographical Information System
R^2	Coefficient of Determination
SCR	Self-Consumption Rate
SOC	State of Charge
TMY	Typical Meteorological Year
TOU	Time-of-Use
TUS	Time Use Survey
XGBoost	eXtreme Gradient Boosting

ABSTRACT

This study suggests a unified framework for the optimal sizing of residential hybrid PV-battery-grid systems specifically designed for solar installers working with unmetered households. The methodology combines three parts to turn simple inputs into useful system designs.

First, a stochastic bottom-up load model will generate 1-minute demand profiles for weekdays and weekends based on household surveys that ask about appliances, occupancy, time-of-use patterns, and lighting. Markov occupancy transitions, scaled appliance probabilities, and lighting that responds to irradiance all will help ensure that the validation is accurate.

Second, machine learning models will employ NASA POWER data from 1981 to 2025 to predict the monthly GHI and temperature at the study site one year in advance. CNN-LSTM, GRU, CNN, XGBoost, and Random Forest will be evaluated using five input features. The best model will be used to give predictions with an uncertainty measure by applying Monte Carlo dropout.

Third, MILP multi-objective optimization will be utilized to balance two objectives: capital costs and self-sufficiency across 576 weighted timesteps (24 representative days). The epsilon-constraint method can be used to obtain three Pareto designs, namely minimum-cost, balanced, and maximum grid independence. The designs will output a PV capacity, battery storage, and inverter specifications.

The structure provides installer-contained solutions, which require minimal input, bridging current state research in HRES with the requirements of residential solar installation in Kenya.

1 INTRODUCTION

1.1 Background

The energy sector in Kenya is facing a growing issue as the rate of electricity demand is higher than the rate of supply. The number of grid connections has grown to 9.7 million in 2023 from 1.5 million in 2010 [1]. The demand is, on the other hand, expected to peak at 5,000 MW by 2030 compared to the small supply of about 4,200 MW [1]. The country has a high interest in clean energy as most of the electricity generated in the country is renewable (90%), with 45 percent of the total production being geothermal. [2].

As a result of the insufficient energy, people are taking matters into their own hands. This is particularly true as they are increasingly turning towards solar PV systems and battery storage solutions amid drastically reduced costs of solar panels, batteries, and overall systems. Increased asset ownership is directly related to the ability to secure alternative sources of power [3]. The country also contributes to this increasing solar market momentum with the favorable policies and the presence of sunlight in 70 percent of its land area [2].

With residential solar adoption gaining momentum, the technology behind it is also growing fast. The logical and natural step is hybrid renewable energy systems (HRES), which integrate photovoltaic panels, advanced lithium-iron phosphate batteries, and smart grid connectivity into a broad-based solution at a household level seamlessly [4,5]. There are solutions that utilize advanced optimization algorithms, such as genetic algorithms and machine learning, to achieve significant cost savings and dependable execution [4].

This intersection of growing national energy demand, increasing popularity of household solar systems, and innovation in hybrid technology is the ideal indicator to target residential research on HRES [1–5].

1.2 Problem statement

Residential solar installers face three design constraints that are critical in undermining the adoption of the systems. First, installers do not have access to the residential household load profile as the majority of installation sites do not have power meters, and clients usually do not accept weeks-long monitoring campaigns to obtain the accurate demand profile [6]. Second, designers use only the historical average sun-hours for PV output design and do not consider the future variation of irradiance between seasons, weather conditions, and climate [6]. Third, the current sizing techniques are project-based, maximizing lifecycle economic indicators to justify projects [7], unlike installer-based objectives that balance between initial capital investment and grid independence to influence client purchasing choices.

In the absence of load profiles, installers set the PV-battery capacity that mismatches household consumption, resulting in an oversized system that wastes money or an undersized system that provides insufficient power to the household [8]. Historical sun-hour averages overlook variations in PV production in the future, leading to continuous underperformance of energy yield. Finally, the lack of clear, installer-friendly solar system options leaves clients confused about what systems to purchase.

Without intervention, customers continue getting subpar investment value while installers lose credibility due to failed systems where design flaws manifest [8]. This creates discontent that spreads through word of mouth, undermining solar technology and stalling market growth when distributed generation is most needed for Kenya's energy security.

1.3 Objectives

The main objective of the study is to develop a comprehensive framework that integrates residential load modelling, long-term solar irradiance and temperature prediction, and multi-objective optimization of hybrid solar systems.

To achieve the main objective, the following specific objectives will be accomplished:

1. To develop and validate a model for creating realistic synthetic residential electricity load profiles.

2. To develop and validate a machine learning model to predict long-term solar irradiance and temperature.

3. To develop and implement a multi-objective optimization approach to sizing hybrid solar systems

1.4 Justification of Study

The framework proposed is based on a realistic input-process-output model, which converts simple household data into valid hybrid solar system designs:

Inputs:

- Household survey lists of appliances by wattages.
- Probabilities of time of use (based on occupancy patterns and appliance use survey)
- Location of the site (latitude/longitude)

Processes:

- Synthetic load profiling for creation of realistic consumption profiles

- Machine learning-based prediction of long-term solar irradiance and temperature
- Multi-objective optimization to balance cost and self-sufficiency / grid independence

Outputs:

Three custom design packages: lowest cost, balanced, and maximum grid independence, all determining the size of PV arrays, battery, the inverter rating and major performance indicators.

This solution directly solves the solar design limitations. Installers replace weeks-long monitoring with instant synthetic load profiles, preventing oversized/undersized failures [6, 8]. The use of long-term solar output prediction removes the pitfalls of historical averages, which makes the forecasts of the yield reliable [6]. The specialization in the needs of installation in optimization providing gives a clear trade-off (cost vs. autonomy) rather than a judicious lifecycle measure [7].

In practice, installers will gain faster evaluations, greater close rates via clear-cut choices, and reduced system issues as a result of better-sized systems. Households are provided with affordable and reliable power in line with their budgetary requirements and grid independence.

1.5 Scope

This study is bounded by the following parameters:

System Scope: Hybrid PV-battery-grid systems for residential applications only, excluding commercial, industrial systems.

Target Users: Solar installers and system designers.

Technical Scope:

- Standard household appliances and consumption patterns
- PV panels
- $LiFePO_4$ batteries
- Hybrid inverters
- Excludes wind, micro-hydro, or other generation sources
- Flat residential electricity tariffs

Data Scope:

- Household survey
- NASA POWER dataset for solar irradiance
- One-month field measurement for model validation

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Geographical Scope: A kenyan household as a case study, with respective climatic conditions and consumption.

2 LITERATURE REVIEW

2.1 Overview

There is a profound research on the use of hybrid renewable energy systems, which incorporate photovoltaic (PV), battery storage, and grid connections in both residential, commercial, and utility-size development and use. [4,7,9]. Studies are concerned with optimization methods that can regulate the technical performance, economic viability, and reliability of the systems under variable renewable generation. Common research themes are multi-objective sizing techniques, uncertainty modeling of PV/load variability, and extensive battery lifecycle management [4].

2.2 Hybrid PV-Battery-Grid Systems Sizing

There are various PV-battery-grid sizing methods in literature. These span single-objective cost minimization to multi-objective frameworks. Classical optimization was used in earlier studies. Fossati et al. applied genetic algorithms to microgrid BESSs sizing for reduction in operating costs [10] while Lu et al. optimized PV-battery setups to reduce demand peaks and minimize grid strain during high-load events [11]. Preliminary works were characterized by analytical approaches, including the definition of energy-to-power ratios as a coefficient by Ashabi et al. [12] to assess the viability of BESS in commercial buildings utilizing minimal time (e.g., usable energy / max power output).

Recent studies apply multi-objective optimization to achieve a balance between economic factors (such as costs), technical reliability (such as outage risks), and environmental benefits (such as emissions cuts). A hybrid Wind/PV/Battery sizing algorithm called the hybrid Grey Wolf-Whale algorithm has been developed by Kapen (2025) [13] to minimize the levelized cost of energy, net present cost (NPC), and loss of power supply probability simultaneously. Chen et al. (2023) [14] utilized NSGA-II to optimize multi-objective PV-BESS energy communities, for maximum self-consumption rate (SCR), minimum payback period, and minimum power transportation loss.

The study by Pirouz et al. [7] characterizes optimization objectives systematically in 40+ studies:

Table 1: HRES Optimization Objectives Taxonomy (from Pirouz et al. 2024)

Category	Objective	Description
Economic	F1: Total Cost	Capital + O&M + Energy Not Supplied penalties (85% studies)
	F2: Load Variance	Peak shaving and load flattening
	F3: IRR	Internal Rate of Return for financial viability
Technical	F4: LPSP	Loss of Power Supply Probability minimization (reliability)
	F5: Emissions	CO_2 reduction and environmental impact
Performance	F6: Market Profits	Energy arbitrage and revenue maximization
	F7: BESS Lifespan	Cycle life and degradation optimization
	F8: Self-Consumption	PV utilization rate enhancement

The latest developments of AI now allow accurate forecasting and optimization of renewable energy systems. Azizi et al. [15] trained CNN, LSTM, GRU, and CNN-LSTM models on monthly data of 1984-2023 to predict global horizontal irradiance (GHI) and temperature. Mohammad et al. [16] systematically reviewed the hybrid CNN-LSTM architecture, reporting better performance in the short, medium, and long-term solar forecasting horizons . Promasa et al. (2025) [17] applied LSTM networks to forecast the residential EV charging demand, followed by optimization of battery scheduling to manage those peaks without overloading the grid. Such studies reveal that AI has the transformative potential to improve the design and operation of a PV-battery-grid system, where forecasting and dynamic optimization under uncertainty are achieved with more accuracy.

HRES optimization requires accurate residential load profiles, yet data acquisition remains challenging. There are two broad methodologies being brought out in literature:

Meter Data / Field Measurements: These measurements offer temporal granularity, measuring high-resolution of existing infrastructure, but necessitate retrofitting in unmetered households. Kapen [13] used this approach for Cameroonian university office loads. Short-term deployments (1-12 months) of power analyzers provide accurate validation quality but are client-averse and are costly to instrument.

Household Surveys: This is a bottom up approach where surveys containing appliance inventories, usage ratios, occupancy rates, and socioeconomic information are used to understand consumption profiles. [18].

2.3 Identified Research Gaps

As observed in the current literature on hybrid renewable energy systems (HRES), there exist four important gaps.

Gap 1: Lack of Practical Residential Load Profiles.

The existing literature is founded on smart meter measurements or costly field measurements [13]. These are not practical in unmetered home or clients who are opposed to long term monitoring. Despite the fact that Leach (2020) assembles the data on appliances to model the energy consumed by cooking [18], no models use such naive inputs (appliance lists, occupancy patterns) to size solar applications.

Gap 2: Reliance on Historical Solar Irradiance Averages.

The optimization models use the past averages of sun-hours that do not take into account the future uncertainties and climatic variations [6, 7]. Even though Azizi et al. demonstrate long-term AI forecasting [15], the key limitations that remain are: there is no validation of alternative location climate, no comparison with baselines (Random Forest, XGBoost) and no prediction intervals with uncertainty analysis. They are necessary to have sound PV yield estimates and risk-based installer choices.

Gap 3: Project vs. Installer Optimization Focus.

Multi-objective sizing targets lifecycle costs, IRR, or LPSP for project financiers [7, 13, 14], not simple comprehensible options such as upfront capital vs. grid autonomy trade-offs to close client sales. No frameworks deliver 3 clear options (cheap, balanced, independent) with PV/battery/inverter specs tailored to household budgets amid Kenya's grid shortages (peak demand 5GW by 2030 vs. 4.2GW supply) [1].

Gap 4: The absence of an integrated sizing framework.

Although studies develop the individual components (load profiling, solar forecasting, optimization), none of the studies incorporate them into a practical all-in-one solution. This discontinuity causes the advanced models not to be applied in practice, delaying development.

These gaps make the adoption of residential HRES stagnant in Kenya, where distributed generation is needed the most for energy security and clean transition objectives [2]. The suggested framework directly satisfies the gaps with viable, validated models that bridge research and installer practice.

3 METHODOLOGY

3.1 Residential Load Modelling

This study develops a stochastic bottom-up household load model. The model is explicitly for residential hybrid PV-battery system sizing. It adapts the CREST demand modeling framework [19] with MECS-CREST modifications [18] for household-specific load applications. The methodology has three steps, namely: (1) model parameterisation through online survey of representative households, (2) Python model implementation, and (3) validation with independent case-study measurements on a monitored household.

3.1.1 Data Collection: Online Survey

The online survey is aimed at 30+ households categorized into consumption archetypes (low <5,kWh/day, medium 5–15,kWh/day, high >15,kWh/day). For each household, the model captures:

- Appliance inventories (type, count, rated power in W)
- Time-of-use (TOU) probabilities and day-type
- Occupancy profiles (active residents by hour, weekday/weekend)
- Lighting configurations (bulb count, wattage, usage patterns)

The survey data of each household directly parameterize the model instance of that household, that is, it produces personalised Markov transition matrices, lists of appliances owned by the household, and lists of TOU/activity probabilities. This swaps the UK TUS statistics of CREST population average with household-specific data [19].

Table 2: Proposed Model vs Predecessors

Aspect	**CREST 1.0B** [19]	**MECS-CREST** [18]	**Proposed Model**
Resolution	1-minute	1-minute	1-minute
Household Rep.	Random synthetic household details per run	Fixed baseline household details per user story	Fixed baseline household details per household
No. of Residents	User defined (1–6)	User defined (cooking diary based)	Survey-specified fixed number
Day	Weekday & Weekend	Single day	Weekday & weekend day × MC diversity × seasonal scaling
Clearness	Loughborough clearness stats for lighting	PVGIS clearness stats for lighting	NASA POWER clearness stats for lighting
Data Sources	UK TUS/ownership stats for average household	Kenyan Cooking diaries + simplified TUS for average household	Specific household TUS/ownership data from surveys tailored for household accuracy
Development Flow	Calibrated to national aggregates	Cooking diaries drive parameters	Household surveys drive parameters
Occupancy	Markov transitions from (UK tables)	Fixed occupancy profile built from cooking diaries	Markov transitions from (Household data)
Appliance Handling	Random ownership/power distributions	User-specified + cooking appliances	Survey-specified ownership
Appliance model	TUS activity probabilities	Simplified TOU + cooking	Survey TUS activity probabilities
Lighting model	Irradiance threshold + random bulbs	Specified LED configurations + irradiance	Survey bulbs + NASA irradiance
Validation Data	Aggregate UK statistics	Illustrative examples only	Independent case-study measurements
Diversity/Scale	Aggregate many runs	Change parameters within/across households	Monte Carlo ensembles
Implementation	Excel/VBA spreadsheet	Modified Excel	Python
Validation Metrics	Calibrated to UK aggregates	Visual examples (Kenya diaries)	MAE, var, P/A ratio vs measured case-study
Output	Single 24h load profile	24h household + aggregate profiles	Weekday + weekend statistical load profiles

3.1.2 Model Development

The proposed 1-minute resolution model is based on the architecture in Table 2

Key components include:

Occupancy Model: First-order Markov chain with survey-derived transition probabilities $P(n_t|n_{t-1})$, where n_t is active residents at minute t.

Appliance Model: For each appliance i, switch-on probability $p_{i,t} = f(\mathrm{TOU}_{i,t}, n_t)$; duration/power drawn from survey distributions.

Lighting Model: $p_{\mathrm{light},t} = g(\mathrm{irradiance}_t, \mathrm{bulbs}, n_t)$ using NASA POWER 1-min clearness index.

Diversity: Monte Carlo ensembles (1000+ runs) with seasonal irradiance scaling will generate weekday and weekend statistical profiles.

3.1.3 Model Validation

One of the residential homes is chosen to be a case-study validation. Occupancy, ratings, and TOU are captured in detail. A meter records the actual 1-min load for a period of 30 days.

Model runs use case-study parameters, producing ensemble profiles. Validation metrics include:

- Mean Absolute Error: $\mathrm{MAE} = \frac{1}{N} \sum |L_{\mathrm{model},t} - L_{\mathrm{meas},t}|$
- Variability ratio: $\sigma_{\mathrm{model}}/\sigma_{\mathrm{meas}}$
- Peak/Average ratio comparison
- 48,h visual profiles (weekday+weekend)

Thresholds: MAE <10%, variability ratio 0.9–1.1. Discrepancies trigger parameter refinement.

3.2 Long-Term Solar Irradiance and Temperature Forecasting

This study develops a physics-informed deep learning ensemble for one-year ahead Global Horizontal Irradiance (GHI) and temperature forecasting at monthly resolution, extending Azizi et al. [15] methodology to Kenyan conditions while incorporating Ghodusinejad et al.'s physics-hybrid recommendations [16]. Five models are benchmarked: CNN-LSTM, GRU, CNN (deep learning), XGBoost, and Random Forest (tree ensembles), all enhanced with Clear Sky Index (CSI) physics feature.

3.2.1 Data Acquisition and Feature Engineering

NASA POWER monthly time series (1981–2025) at Nairobi (1.2921°S, 36.8219°E)

- **Targets**:
  - ALLSKY_SFC_SW_DWN (GHI, kWh/m^2/day)
  - T2M (2m air temperature, °C) enditemize
  - **Baseline (4 meteorological)**:
    - GHI_t (current month global horizontal irradiance)
    - T2M_t (2m temperature)
    - RH2M_t (2m relative humidity, %)
    - PS_t (surface pressure, kPa)
  - **Physics-informed enhancement (1)**:

* **Clear Sky Index (CSI)$_t$**:

$$\text{CSI}_t = \frac{\text{GHI}_{\text{measured},t}}{\text{GHI}_{\text{clear-sky},t}} \tag{1}$$

where $\text{GHI}_{\text{clear-sky},t}$ computed via Ineichen clear-sky model (linke turbidity $T_L = 3.5$ for Nairobi). Captures cloud attenuation physics missing from raw GHI.

Input matrix: 12 months $\times$ 5 features = **60 inputs per prediction**

3.2.2 Monthly Rolling Forecast

Models predict one month ahead using prior 12 months, rolled forward iteratively for complete one year ahead coverage:

1. **Training**: 1981–2024 monthly data
2. **Input window**: (2025) Months $t-11$ to t (12 months $\times$ 5 features = 60 inputs)
3. **Target**: Month $t+1$ (GHI, temperature)
4. **Roll-forward**: Jan 2026 through Dec 2026 (12 predictions)

Example: Jan–Dec 2025 data $\rightarrow$ Predict Jan 2026 GHI/temp; Feb 2025–Jan 2026 $\rightarrow$ Feb 2026, etc.

3.2.3 Model Formulations

Five architectures process the 60-input matrix (12 months $\times$ 5 features: GHI, T2M, RH2M, PS, CSI) to predict next-month GHI and temperature:

CNN-LSTM (Hybrid Sequential Processing): Combines CNN feature extraction with LSTM temporal memory. The CNN layer applies 32 filters (kernel size 3) across the 12-month window to detect local seasonal patterns, which are then fed sequentially into a 2-layer LSTM (64 hidden units per layer) maintaining month-to-month dependencies:

$$\mathbf{h}_t = \text{LSTM}\big(\text{ReLU}\big(\text{Conv1D}(\mathbf{X}_{t-11:t})\big), \mathbf{h}_{t-1}\big) \tag{2}$$

GRU (Efficient Sequential Memory): Simplified LSTM architecture using 2 gating mechanisms (update gate z_t, reset gate) instead of LSTM's 3 gates, reducing parameters by 33% while preserving long-term memory capability:

$$\mathbf{h}_t = (1 - z_t) \odot \mathbf{h}_{t-1} + z_t \odot \tilde{\mathbf{h}}_t \tag{3}$$

CNN (Parallel Pattern Recognition): Purely convolutional approach processes the entire 12-month window simultaneously without recurrent connections. 32 filters (kernel=3) slide across all months to extract seasonal patterns directly:

$$\mathbf{h}_t = \text{ReLU}\big(\text{Conv1D}(\mathbf{X}_{t-11:t}, 32 \text{ filters, kernel} = 3)\big) \tag{4}$$

Tree Baselines (Non-Temporal):

- **XGBoost**: Gradient boosting with 500 trees (max depth 6), where each tree sequentially corrects errors of previous trees through second-order gradient optimization
- **Random Forest**: Bagging ensemble of 500 independent decision trees (max depth 6), final prediction via majority vote/average

Model Comparison:

3.2.4 Data Split and Evaluation

Standard 60/20/20 temporal split:

Final 2026 forecasting uses models retrained on complete 1981–2025 dataset.

Table 3: Model Architectures Comparison

Model	Time Patterns	Training Speed	Best For
CNN-LSTM	Yes	Slow	Long dependencies
GRU	Yes	Medium	Fast memory
CNN	Yes	Fastest	Seasonal patterns
XGBoost	No	Fast	Feature math
Random Forest	No	Fast	Stability

Table 4: Data Split for Model Development

Set	Years	Months	Purpose
Training	1981–2013	396 (60%)	Parameter learning
Validation	2014–2018	60 (20%)	Hyperparameter tuning
Test	2019–2025	84 (20%)	Final evaluation
Forecast	2026	12	PV sizing

3.2.5 Evaluation and Model Selection

Performance will be checked on the basis of typical time-series measures against persistence bottom line (foresee next month = present-day month):

Metrics:

- R^2: Fraction of variance explained (perfect=1.0)
- MAE: Mean Absolute Error (W/m^2 for GHI, °C for T2M)
- vs Persistence: Improvement over naive benchmark
- **Prediction Intervals**: 80%/95% uncertainty bands via Monte Carlo dropout (deep learning) or quantile regression (XGBoost)

Uncertainty Quantification (Best Model):

- **Test Set Residuals**: Empirical $\pm 2\sigma$ bounds from validation residuals

- **MC Dropout**: 100 forward passes through best deep learning model at inference for uncertainty bands
- **Output**: Monthly GHI/temp forecasts $\hat{y}_m \pm \sigma_m$ for robust PV sizing

Model ranking factors:

- Test set R^2 (primary)
- Uncertainty calibration (80% coverage on test set)
- Training stability (no overfitting)
- Computational efficiency (training + inference)

Best single model selected via weighted ranking for 2026 forecasting with uncertainty estimates.

3.2.6 PV Sizing Integration

2026 ensemble GHI drives PV yield estimation for hybrid system sizing optimization.

3.3 Multi-Objective Sizing Optimization

This study uses a MILP-based multi-objective optimization framework for optimal sizing of hybrid PV-battery-grid systems. Using 24 representative days (12 months $\times$ weekday/weekend = 576 hourly timesteps) from the household demand model and PV generation forecasts from the solar irradiance/temperature model, three Pareto-optimal designs are generated: cheapest, balanced, and high-autonomy.

3.3.1 System Model Formulation

The hybrid system comprises:

- **PV Array**: Converts solar irradiance forecasts G_t (W/m^2) and temperature T_t (°C) into electrical energy using linear approximation:

$$P_t^{\mathrm{PV}} = P_{\mathrm{PV}} \cdot \frac{G_t}{G_{\mathrm{STC}}} \cdot [1 + \gamma(T_t - T_{\mathrm{STC}})] \tag{5}$$

  where P_{PV} is PV capacity (kW), $G_{\mathrm{STC}} = 1000$ W/m^2, $T_{\mathrm{STC}} = 25$°C, $\gamma = -0.004$ °C^{-1}.

- **LiFePO$_4$ Battery Storage**: Stores excess PV energy with capacity E_{bat} (kWh) and power rating P_{bat} (kW). SOC dynamics follow:

$$\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \eta_c P_t^c \Delta t - \frac{P_t^d}{\eta_d} \Delta t \tag{6}$$

  with $0.2 \leq \mathrm{DOD} \leq 0.9$, $\eta_c = 0.95$, $\eta_d = 0.98$.

- **Grid Connection**: 240 V AC, 50 Hz; supplies deficits or absorbs surplus at a flat tariff.

- **Inverter**: Hybrid rating P_{inv} (kW) satisfies $P_{\mathrm{inv}} \geq \max(P_{\mathrm{PV}}, P_{\mathrm{bat}})$.

3.3.2 Decision Variables

Three continuous decision variables are optimized:

$$x = [P_{\mathrm{PV}}, E_{\mathrm{bat}}, P_{\mathrm{inv}}] \quad \text{(kW, kWh, kW)} \tag{7}$$

3.3.3 Objective Functions

The optimization balances two competing goals over T=576 timesteps (24 representative days) with time-weighted objectives representing full-year performance.

Cost Minimization (f_1): Upfront Capital Investment The first objective minimizes total system cost:

$$f_1(x) = c_{\mathrm{PV}} P_{\mathrm{PV}} + c_{\mathrm{bat}} E_{\mathrm{bat}} + c_{\mathrm{inv}} P_{\mathrm{inv}} + c_{\mathrm{BOS}} \tag{8}$$

Self-Sufficiency Maximization (f_2): Energy Independence The second objective maximizes the weighted fraction of household energy served by PV+battery:

$$f_2(x) = -\frac{\sum_{t=1}^{576} E_t^{\text{served}} \cdot w_t}{\sum_{t=1}^{576} L_t \cdot w_t} \tag{9}$$

where w_t = day weight representing full-year equivalence.

MILP Linearization Nonlinear $\min(P_t^{\text{PV}} + P_t^d, L_t)$ becomes linear constraints:

$$E_t^{\text{served}} \leq P_t^{\text{PV}} + P_t^d \tag{10}$$

$$E_t^{\text{served}} \leq L_t \tag{11}$$

3.3.4 Complete Optimization Problem (T=576 Timesteps)

Decision Variables

- $P_{\text{PV}}, E_{\text{bat}}, P_{\text{inv}}$: System sizes
- $P_t^{\text{PV}}, P_t^c, P_t^d, \text{SOC}_t$: Hourly operations (t=1...576)
- $u_t^c, u_t^d \in \{0, 1\}$: Binary charge/discharge switches

Key Constraints

$$L_t = P_t^{\text{PV}} + P_t^d + P_t^{\text{grid}} + E_t^{\text{ns}} \tag{12}$$

$$P_t^{\text{PV}} \leq P_{\text{PV}} \cdot \frac{G_t}{G_{\text{STC}}} \tag{13}$$

$$P_t^c \leq P_{\text{bat}} u_t^c, \quad P_t^d \leq P_{\text{bat}} u_t^d \tag{14}$$

$$u_t^c + u_t^d \leq 1 \tag{15}$$

Days of Autonomy Constraint: Minimum battery backup for grid outages:

$$E_{\text{bat}} \geq D \cdot \bar{L}_{\text{daily}} \cdot \frac{1}{\eta_d \cdot \text{DOD}} \tag{16}$$

where $D = 1$ day autonomy, $\bar{L}_{\text{daily}}$ = weighted average daily load ($\sum L_t \cdot w_t/365$), $\eta_d = 0.98$, DOD=0.7 (80% usable capacity).

This ensures the high-autonomy design provides at least 1 full day backup during grid outages while optimizing economics across the Pareto front.

ε-Constraint Method

$$\min f_1(x) \quad \text{s.t.} \quad f_2(x) \geq \epsilon_i, \quad i = 1, \ldots, 50 \tag{17}$$

3.3.5 Representative Day Weighting (365-Day Equivalence)

24 representative days (12 months × weekday/weekend) weighted for full-year accuracy:

Table 5: Day Weights w_t for T=576 Timesteps ($\Sigma w_t = 365$ days)

Month	Days	Weekday (w_t)	Weekend (w_t)	Timesteps
Jan	31	22.14	8.86	1-48
Feb	28	20.00	8.00	49-96
Mar	31	22.14	8.86	97-144
Apr	30	21.43	8.57	145-192
May	31	22.14	8.86	193-240
Jun	30	21.43	8.57	241-288
Jul	31	22.14	8.86	289-336
Aug	31	22.14	8.86	337-384
Sep	30	21.43	8.57	385-432
Oct	31	22.14	8.86	433-480
Nov	30	21.43	8.57	481-528
Dec	31	22.14	8.86	529-576

w_t = days_in_month × daytype_fraction (5/7 weekday, 2/7 weekend)

3.3.6 Data Integration

- **Load data**: Hourly weekday/weekend profiles from 1-min demand model
- **PV data**: Monthly 2026 GHI/temp forecasts scaling NASA TMY daily shapes

- **Parameters**: $LiFePO_4$ (DOD=80%), KPLC flat tariff
- **Solver**: PuLP/Gurobi (ε-constraint, 50 points, T=576)

3.3.7 Design Selection

Three designs are extracted from the Pareto front $\mathcal{P}$ using the following selection rules:

- **Cheapest**: $\min_{p \in \mathcal{P}} f_1(p)$ – Targets budget-limited customers.
- **Balanced**: Knee point identified via the elbow method. Suited for most households.
- **High Autonomy**: $\max_{p \in \mathcal{P}} f_2(p)$. Designed for customers prioritizing energy independence.

KPIs: Cost (KES), Self-sufficiency (%), P_{PV}, E_{bat}, P_{inv}.

4 EXPECTED RESULTS

The following results are expected at the end of this research:

1. A validated probabilistic load model capable of generating realistic high-resolution synthetic demand profiles for the case-study household.

2. A tested machine-learning model that provides accurate long-term GHI forecasts and corresponding PV energy yield time series, including basic uncertainty estimates for the study location.

3. A validated MILP-based sizing method generating Pareto-optimal PV–battery–grid system sizes summarized into three practical design options (cheapest, balanced, high-autonomy) with quantified objective indicators.

5 BUDGET

Table 6 shows the project budget breakdown.

Table 6: Budget Breakdown

Item	**Description**	**Cost (KES)**
Laptop	Mid-range laptop for simulations/ML models	80,000
Power meter	Single-phase DIN-rail meter for household load monitoring	40,000
Transport	Matatu/taxi fares for field visits	3,000
Misc. expenses	Printing, data bundles, stationery	7,000
Total		**130,000**

6 WORK SCHEDULE

The research is expected to take approximately twelve (12) months. This time will be distributed among the various activities as shown in Figure 1. Some activities will run concurrently.

Activity	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb
Literature review	●	●	●	●	●	●	●	●	●	●	●	●
Proposal & defense	●	●	●									
Online survey & data processing		●	●									
Load model development			●									
Model validation & Case study evaluation			●	●	●							
DL irradiance model training				●	●	●	●					
PV sizing optimisation					●	●	●	●				
Framework validation & metrics							●	●				
Results analysis							●	●	●			
Thesis writing							●	●	●	●	●	
Journal paper drafting & submission									●	●	●	
Thesis revision & defense										●	●	●

Figure 1: Project Work Plan (Mar 2026 - Feb 2027)

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